Arrow Cage
OLLIF
Surgical Technique Guide

Lumbar Interbody Fusion System (OLLIF)

K173947

MM-01-1 REV C
2023-03-10

Arrow
Lumbar Interbody Fusion System
(OLLIF)

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Disclaimer:

The surgical technique shown is for illustrative purposes only. The technique(s) actually employed in each case will always depend upon the medical judgment of the surgeon before and during surgery as to the best mode of treatment for each patient. Please reference the 510(k) or IFU statement for additional information and a complete list of intended indications, warnings, precautions, and other medical information.

Instrument Guide

MP214-3.0/1

Disposable Monopolar Probe (with Sleeve) from Technomed

01-9080-18U

Guide Wire - 1.4 mm X 18 in [Unthreaded]

01-9080-18T

Guide Wire - 1.4 mm X 18 in [Threaded]

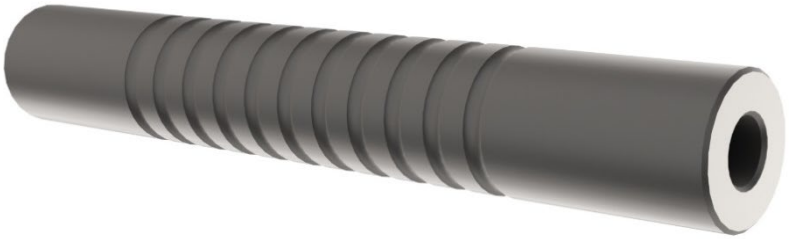
Instrument Guide

01-01
Dilator

01-100
1/4" Square Quick Connect Palm Handle

01-02
Access Portal

Instrument Guide



01-10
Impactor

01-11
Mallet

01-03
Disc Drill

Instrument Guide
Expanding Paddle Shaper

01-04-1
Shaper – Body

01-04-2
Shaper – Knob

01-04-3
Shaper – Blades

Instrument Guide

01-05
Pituitary Rongeur

01-07
Articulating Curette

Instrument Guide
Flexible Curette

01-06-1
Flexible Curette – Inner Shaft

01-06-2
Flexible Curette – Outer

01-06-3
Flexible Curette – Knob

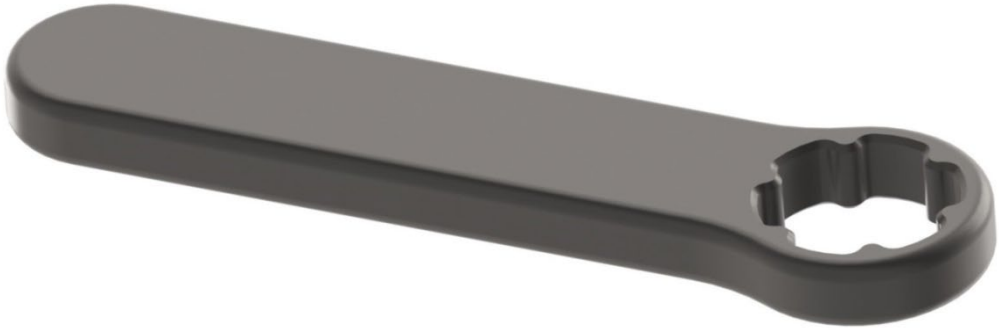
01-06-4
Flexible Curette – Blade

01-101
Mini AO Palm Handle

Instrument Guide

01-08
Graft Delivery Tube

01-09
Graft Tamp



01-12
Wrench

Instrument Guide
OLLIF Inserter

01-00-1
OLLIF Inserter - Outer

01-00-2
OLLIF Inserter - Inner

01-LLHH
Arrow Implant

Instrument Guide

Table of Components

PN	DESCRIPTION	Quantity per set
01-00	OLLIF Inserter [Assembly]	N/A
01-00-1	OLLIF Inserter – Outer	2
01-00-2	OLLIF Inserter – Inner	2
01-01	Dilator	1
01-02	Access Portal	1
01-03	Disc Drill	1
01-04	Shaper [Assembly]	N/A
01-04-1	Shaper – Body	1
01-04-2	Shaper – Knob	1
01-04-3	Shaper – Blades	2
01-05	Pituitary Rongeur	1
01-06	Flexible Curette [Assembly]	N/A
01-06-1	Flexible Curette – Inner Shaft	1
01-06-2	Flexible Curette – Outer	1
01-06-3	Flexible Curette – Knob	2
01-06-4	Flexible Curette – Blade	3
01-07	Articulating Curette	1
01-08	Graft Delivery Tube	1
01-09	Graft Tamp	1
01-10	Impactor	1
01-11	Mallet	1
01-12	Wrench	1
01-100	¼” Square Quick Connect Palm Handle	1
01-101	Mini AO Palm Handle	1
01-9080-18T	Guide Wire – 18” Threaded	20
01-9080-18U	Guide Wire – 18” Unthreaded	20
MP 214-3.0/1	Neuromonitoring Probe with Sleeve	1 per Sterile Package

Instrument Guide

Table of Implants

Part Number	Length (mm)	Height (mm)	Color
01-2708	27	08	Dark Green
01-2709	27	09	Purple
01-2710	27	10	Dark Blue
01-2711	27	11	Copper
01-2712	27	12	Light Gold
01-2713	27	13	Light Green
01-2714	27	14	Light Magenta
01-2715	27	15	Light Blue
01-3008	30	08	Dark Green
01-3009	30	09	Purple
01-3010	30	10	Dark Blue
01-3011	30	11	Copper
01-3012	30	12	Light Gold
01-3013	30	13	Light Green
01-3014	30	14	Light Magenta
01-3015	30	15	Light Blue
01-3308	33	08	Dark Green
01-3309	33	09	Purple
01-3310	33	10	Dark Blue
01-3311	33	11	Copper
01-3312	33	12	Light Gold
01-3313	33	13	Light Green
01-3314	33	14	Light Magenta
01-3315	33	15	Light Blue

Surgical Technique

The surgical approach for the OLLIF technique is based on the anatomic structure known as Kambin's Triangle. The implant and instruments were created to take advantage of the safe, minimally disruptive pathway that leads through Kambin's Triangle to the lumbar disc spaces. This guide will demonstrate the surgical technique for Kambin's Triangle access.

Kambin's Triangle is defined by the combination of the anatomic structure of the lumbar neural foramen. On the inferior leg of the triangle is the superior endplate of the caudal vertebral body. The medial leg of the triangle is bordered by the superior articular process of that same vertebral body and the hypotenuse of Kambin's Triangle is defined by the nerve root exiting from the foramen.

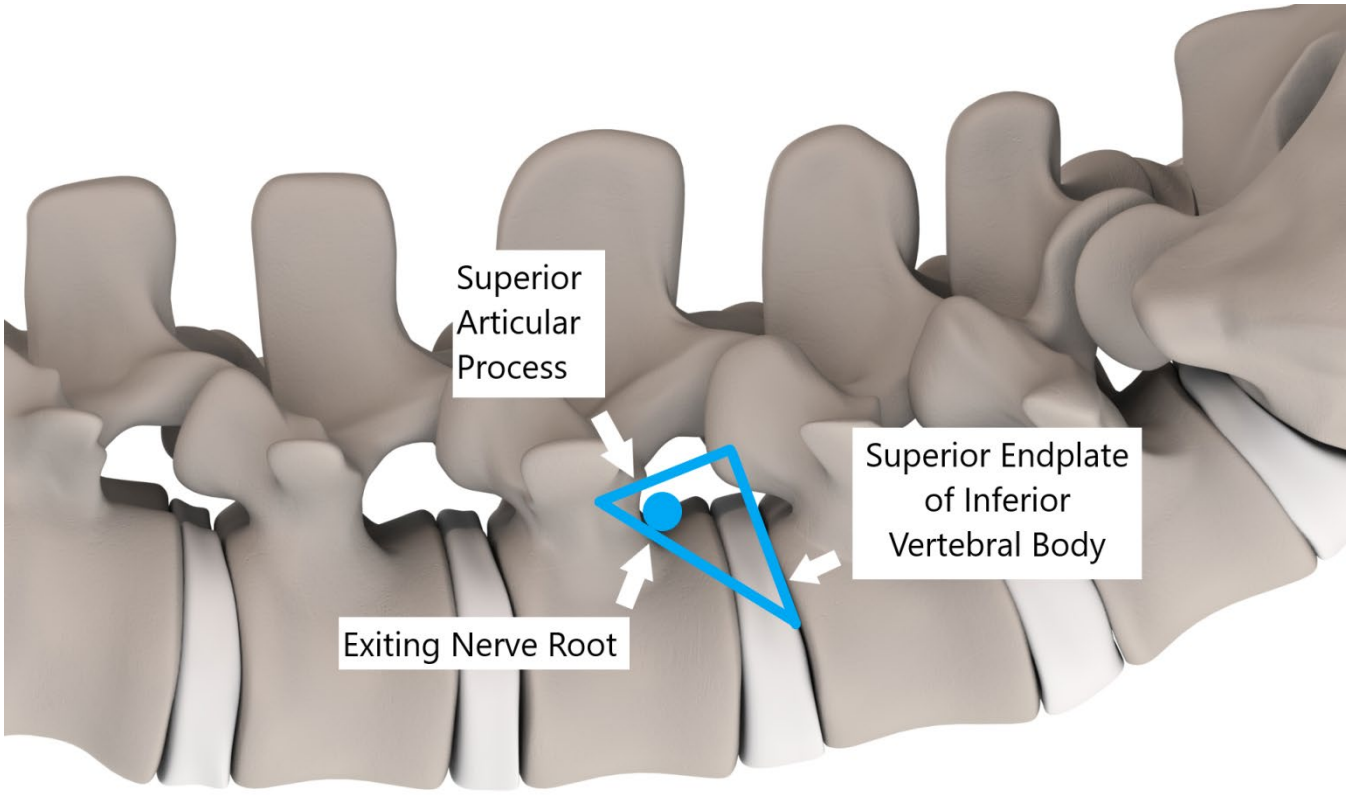


Figure 1

Surgical Technique

How The OLLIF Approach Differs From Other Surgical Approaches

TLIF– the Trans-Facet Lumbar Interbody Fusion approach removes some or all of the facet on the side of approach, which allows for an approach to the disc space at an angle that is more vertical than the OLLIF approach. While the TLIF approach removes bone from the facet structure, the OLLIF approach preserves the facet structure.

Considerable scarring is often seen in TLIF approaches and can lead to adhesions to the dura and the exiting nerve roots. The OLLIF technique is intended to minimize disruption to the adjacent tissues and reduce the occurrence of scarring, and muscle atrophy.

XLIF/DLIF – the trans-psoas lateral approach enters the disc space anterior to the structure referred to as Kambin's Triangle, and must pass through the psoas muscle. The nerves of the lumbar plexus are near the pathway of the trans-psoas approach. Due to the large cage profile, the trans-psoas approach requires significant retraction and dilation of the psoas muscle. This dilation may lead to injury of the nerves in the lumbar plexus.

The OLLIF approach requires minimal dilation and retraction as it passes through Kambin's Triangle because it is completely posterior to the iliopsoas.

Targeting

Preparing for the skin incision can be carried out by use of fluoroscopic imaging. A critical step is to confirm that the imaging is oriented correctly on the target level. Confirm that the A/P image is properly centered and is showing the correct image of the vertebral body. This can be achieved by laying a guide wire or other radiopaque instrument on the midline to confirm that the A/P image shows the pedicles are equally spaced on each side of the midline. Mark this line.**(Figure 2- Red Line)**

Next, the trajectory of the target disc should be identified, and a mark placed across the skin to locate the center of the disc. **(Figure 2- Orange Line)**

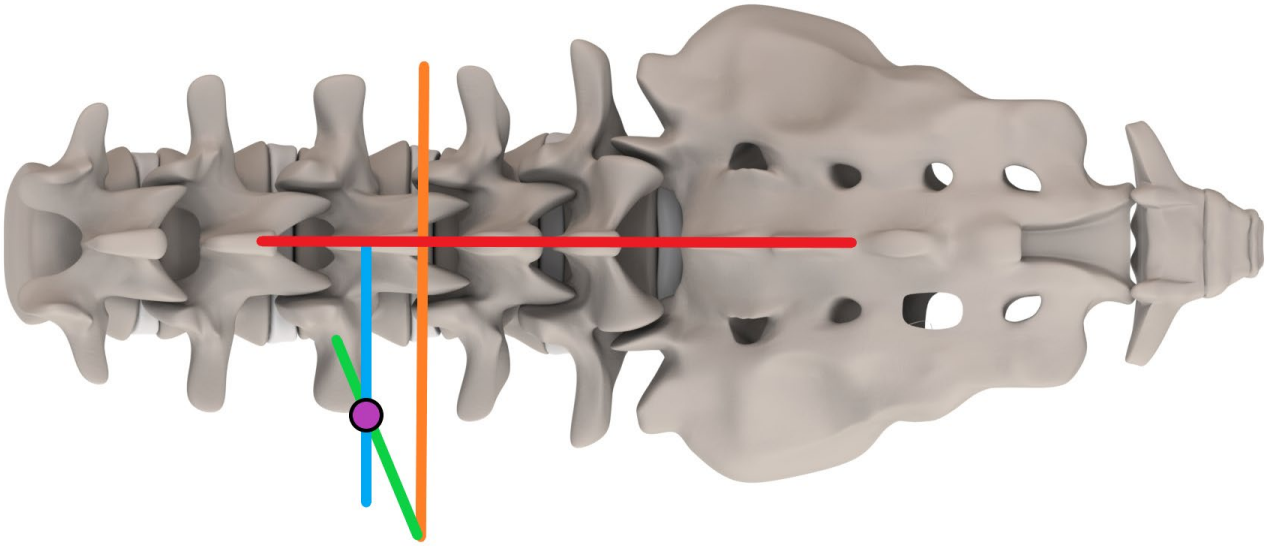


Figure 2

Targeting

Moving to a lateral view, an opaque object should be placed to estimate the center of the disc, between the endplates, with the tip of the instrument located at approximately the center of the disc. The endplates must be perfectly aligned in the lateral view.

Mark the skin to record the angle of the disc and make that mark extend over the posterior flank. **(Figure 3- Green Line)**

The distance from the center of the disc to the apex of the curve of the flank is measured, then this distance is transferred in the same plane from the midline of the spine.

This is the same distance that will be used to measure from the midline to the appropriate incision point. This will create a theoretical right triangle with the end of the hypotenuse in the center of the disc. **(See Figure 4)**

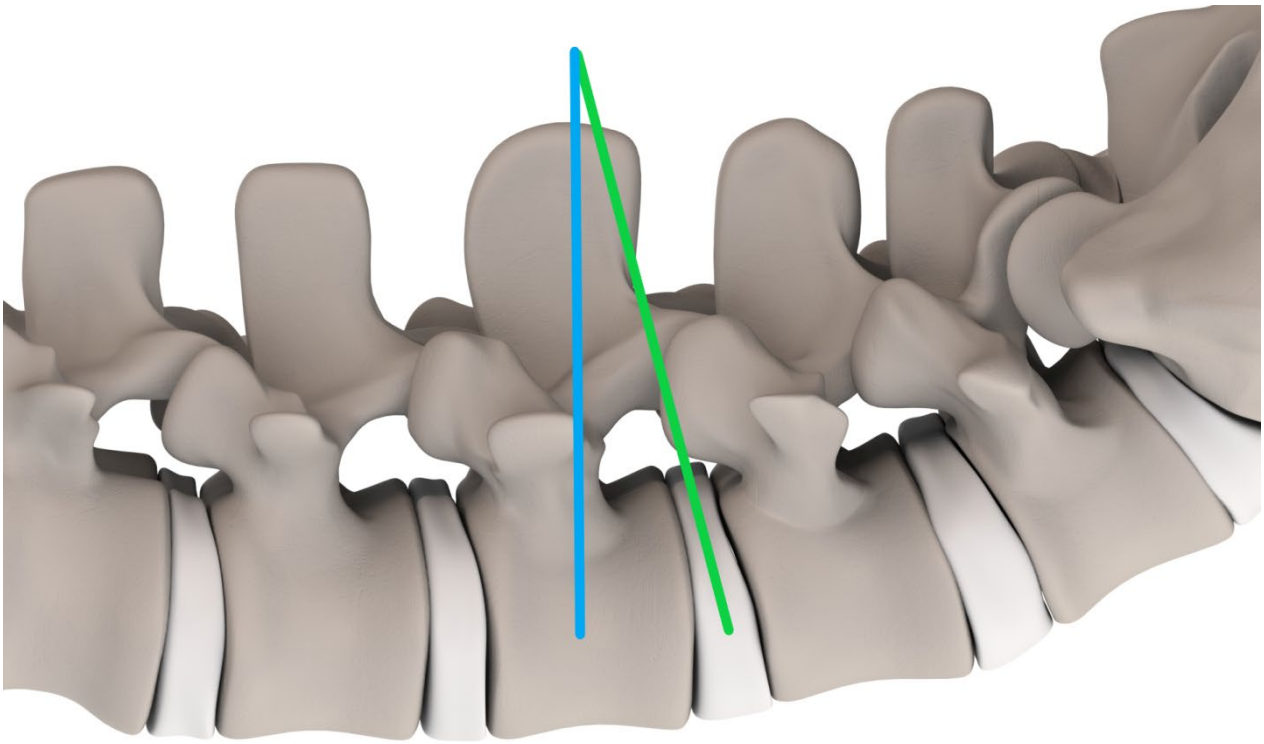


Figure 3

Targeting

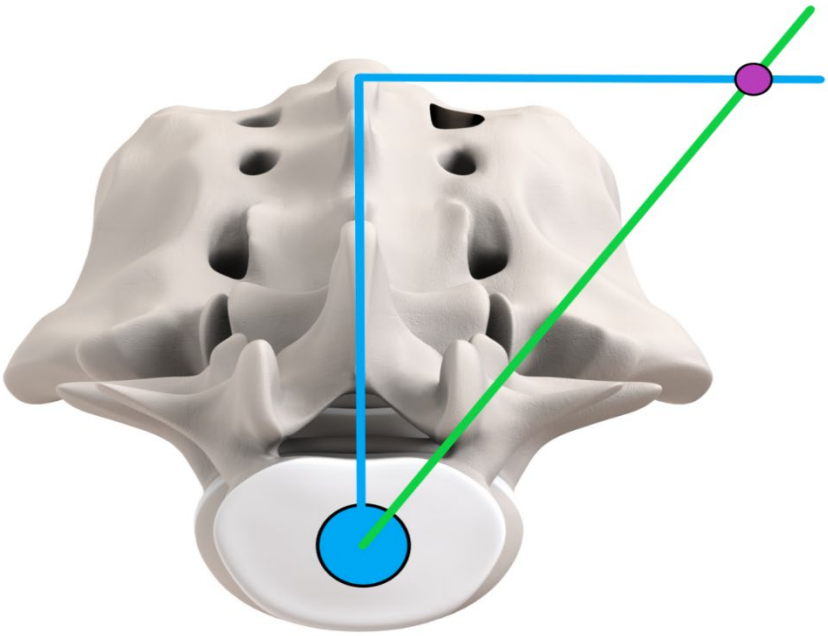


Figure 4

To confirm the incision point is in a place that will lead to Kambin's Triangle without obstruction, a spinal needle may be placed to confirm the trajectory. After confirming the trajectory, make a 10 mm incision in the skin and through the lumbar fascia.

Targeting

Place the **neuromonitoring probe** through the incision. Approach the neural foramen and target the inferior pedicle. Palpate along the surface of the pedicle until the tip of the **probe** is located at the inferior edge of the foramen. The final placement of the **probe** is correct when the tip is between the medial and lateral aspect of the pedicle and in the inferior section of the neural foramen. **(Figure 5,6 and 7)**

Do not cross the medial border of the pedicle as seen in the A/P fluoro view until the tip of the **neuromonitoring probe** is anterior to the posterior aspect of the vertebral body in the lateral fluoro view. This will prevent the trajectory from traversing the spinal canal.

Use neurostimulation to confirm that the tip of the **probe** is safely located with respect to the exiting nerve root at the target level.

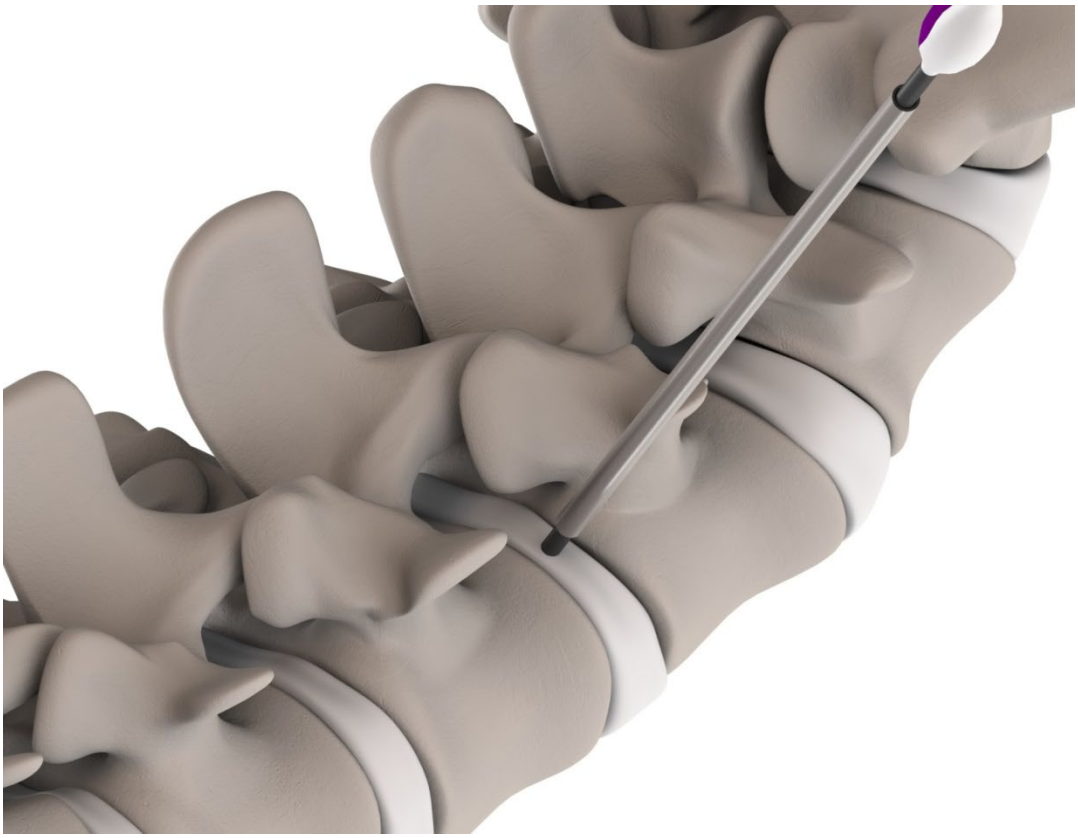


Figure 5

Targeting

By a combination of fluoroscopic imaging and neuromonitoring, the safe pathway to the neural foramen and the disc can be achieved.

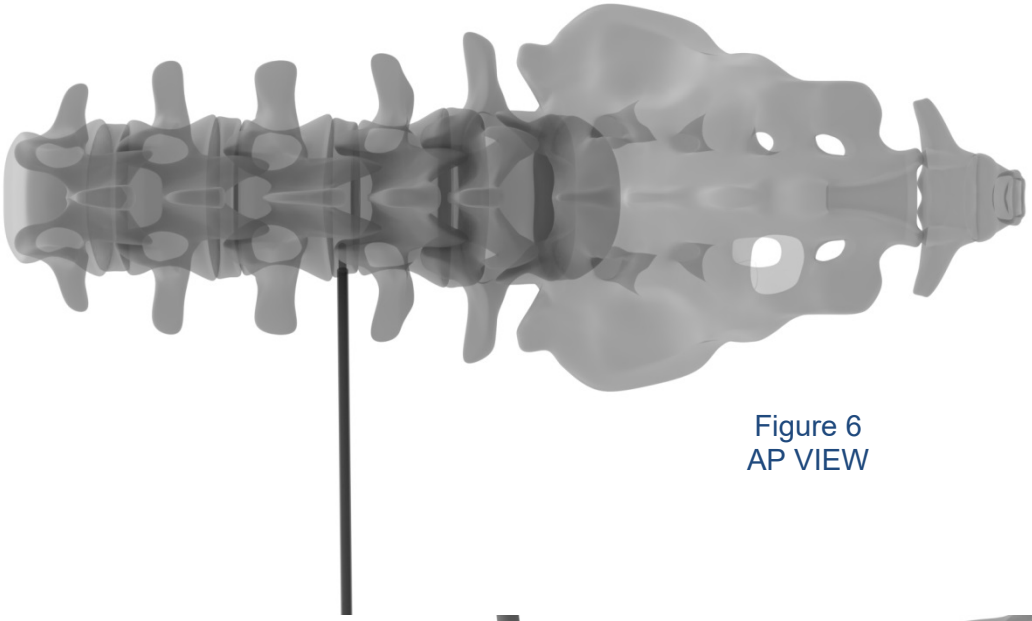


Figure 6
AP VIEW

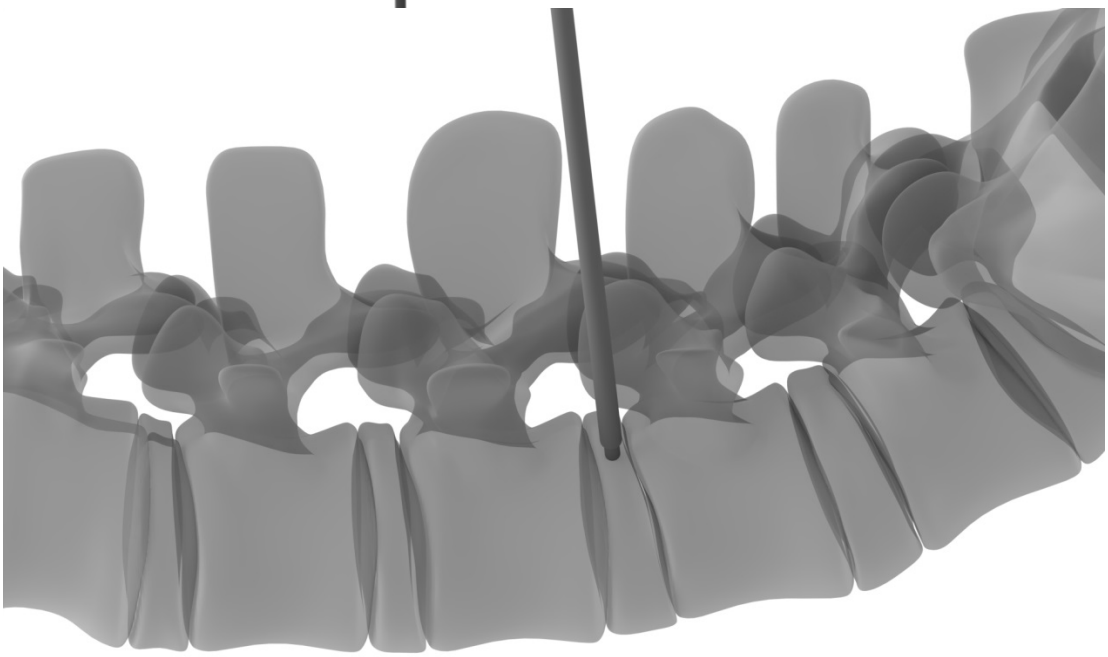


Figure 7
LATERAL VIEW

Targeting

After completing the approach to the disc, press the **neuromonitoring probe** sleeve forward onto the anulus of the disc and remove the **monopolar neuromonitoring probe**. **(Figure 8)**

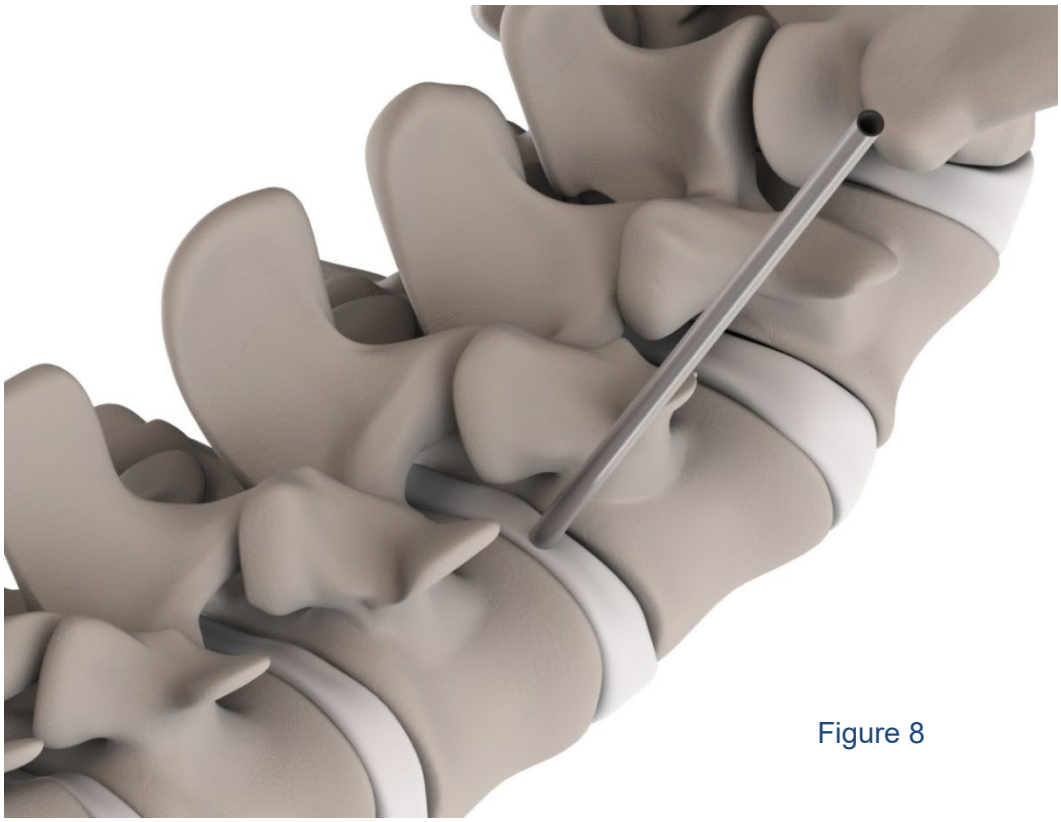


Figure 8

Place an 18” **guide wire** through the **neuromonitoring probe sleeve** and into the center of the disc **(Figure 9).** Confirm the placement of the **guide wire** via fluoroscopy **(see figure 10 and figure 11**).

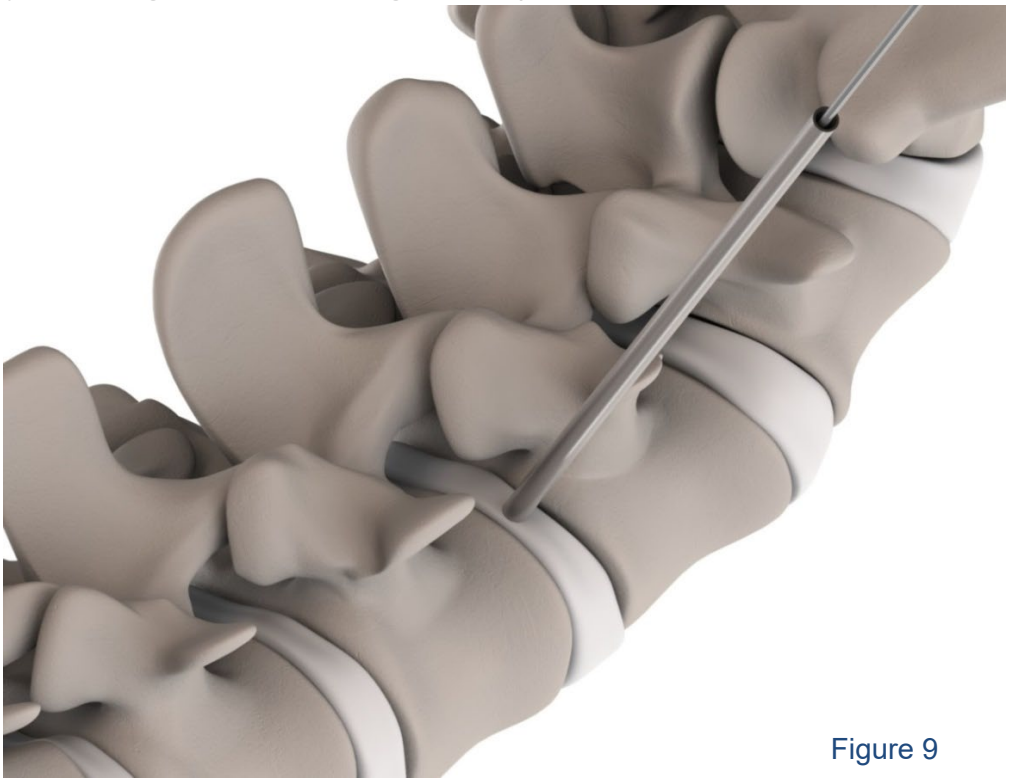


Figure 9

Targeting

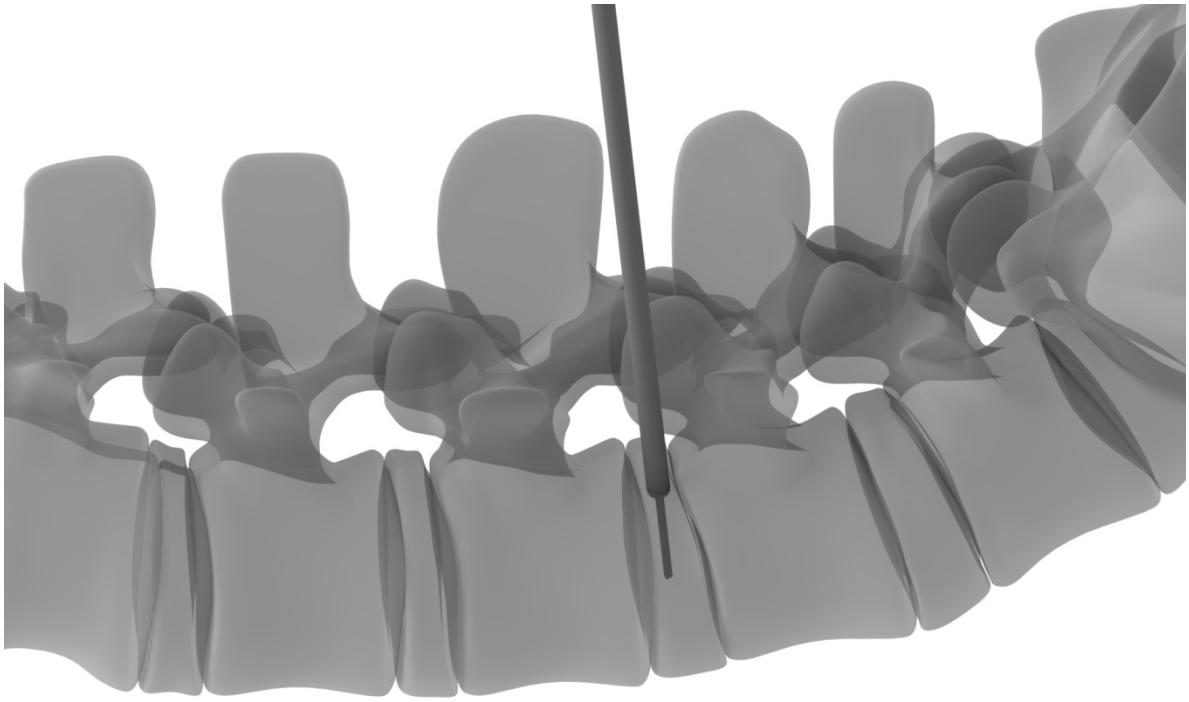


Figure 10

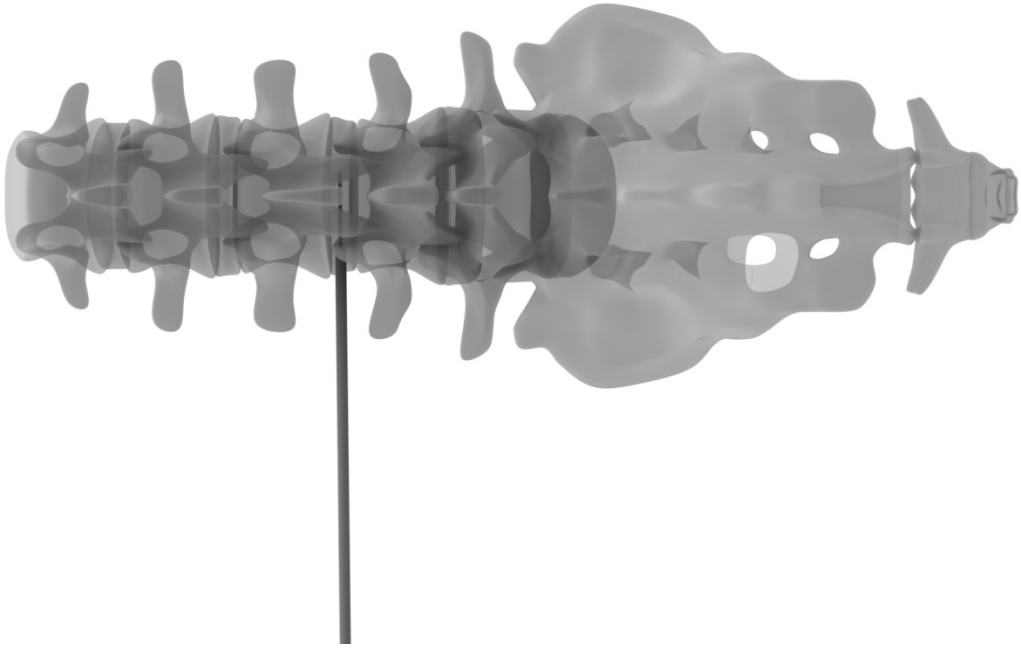


Figure 11

Targeting

After the pathway to the disc has been safely established, the rest of the surgery can be done with EMG and SSEP Monitoring.
Place the **dilator** over the **guide wire**, and confirm fluoroscopically that the tip passes up to center of the disc on A/P and lateral images.

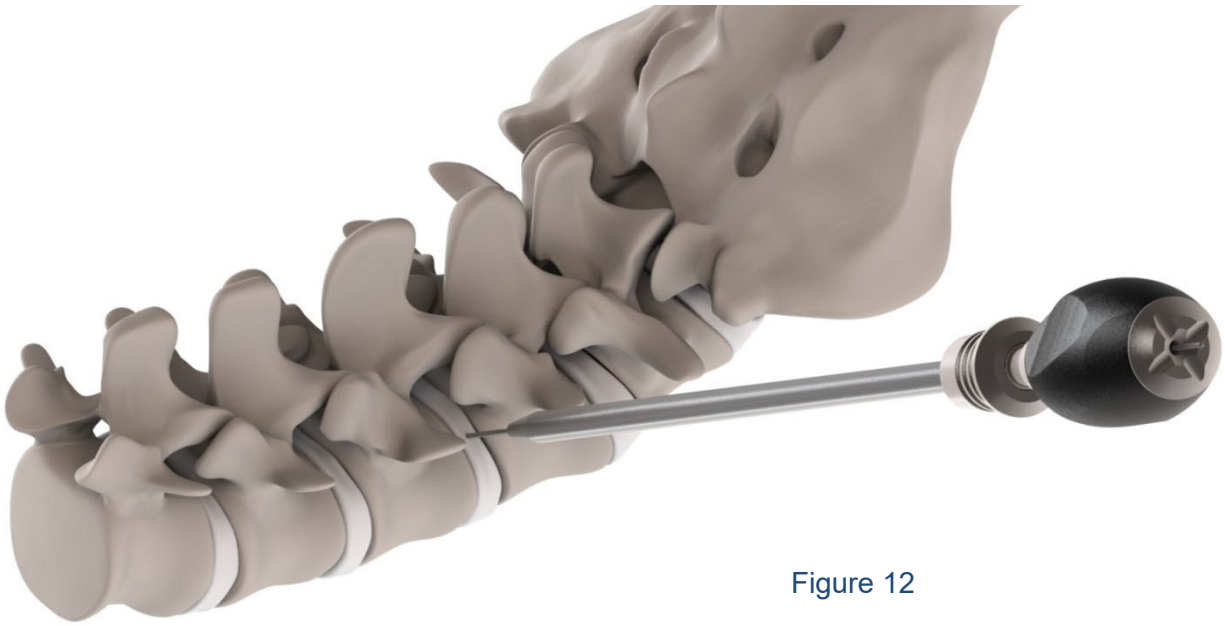


Figure 12

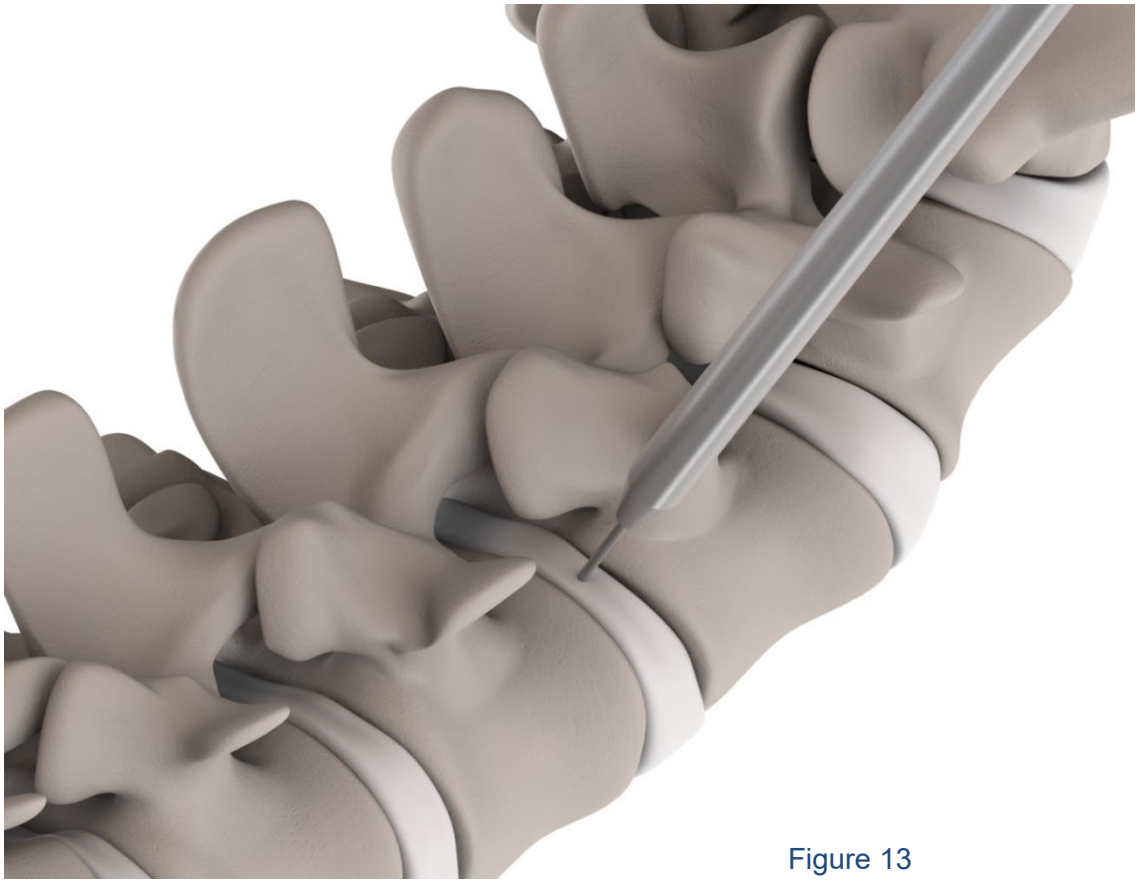


Figure 13

Targeting

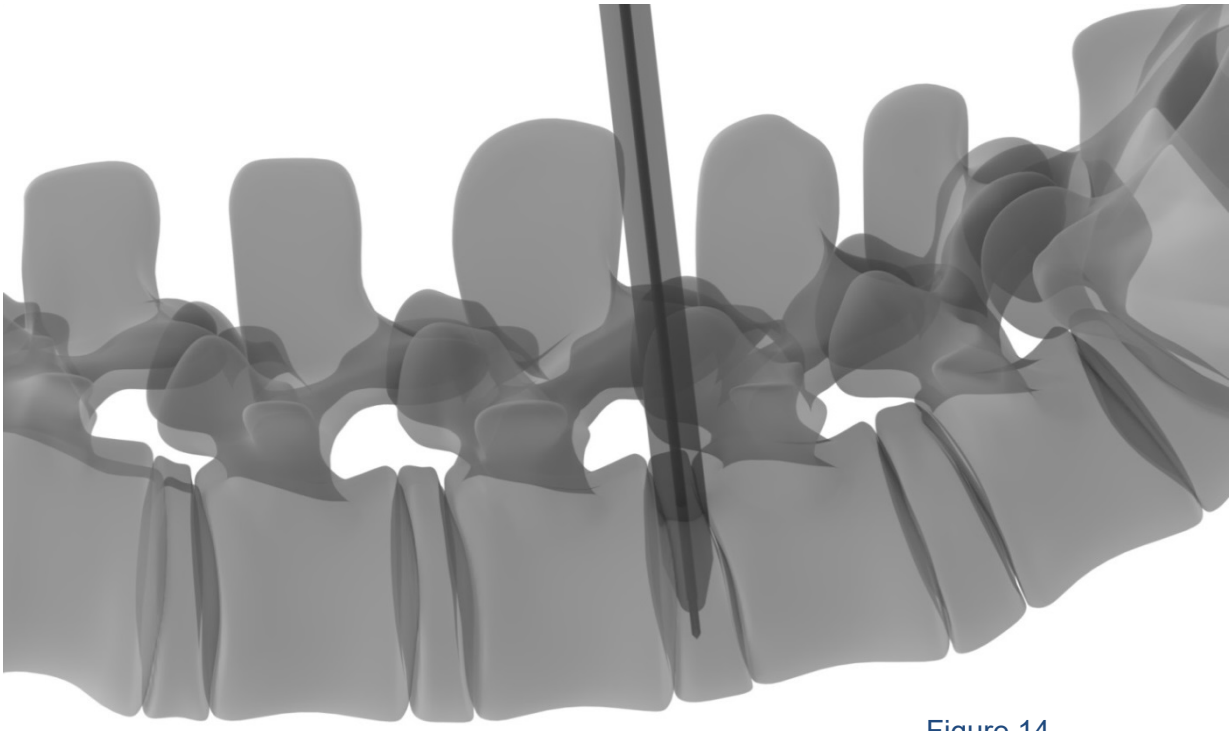


Figure 14

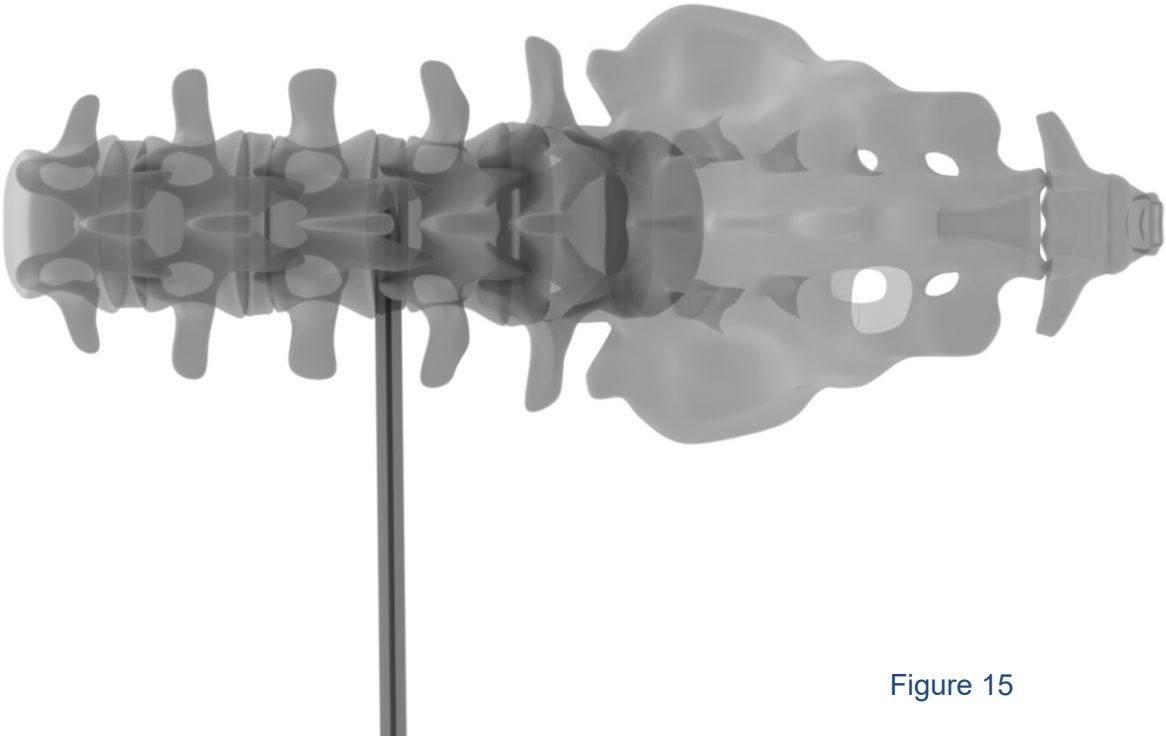


Figure 15

After confirming the **dilator** is 1 cm into the disc space, remove the **guide wire**. After the tip of the **dilator** has reached the midline, remove the **handle**.

Targeting

Advance the **access portal** over the **dilator** until the end of the **portal** is secured inside the disc space. The **impactor** can be used to facilitate this placement.

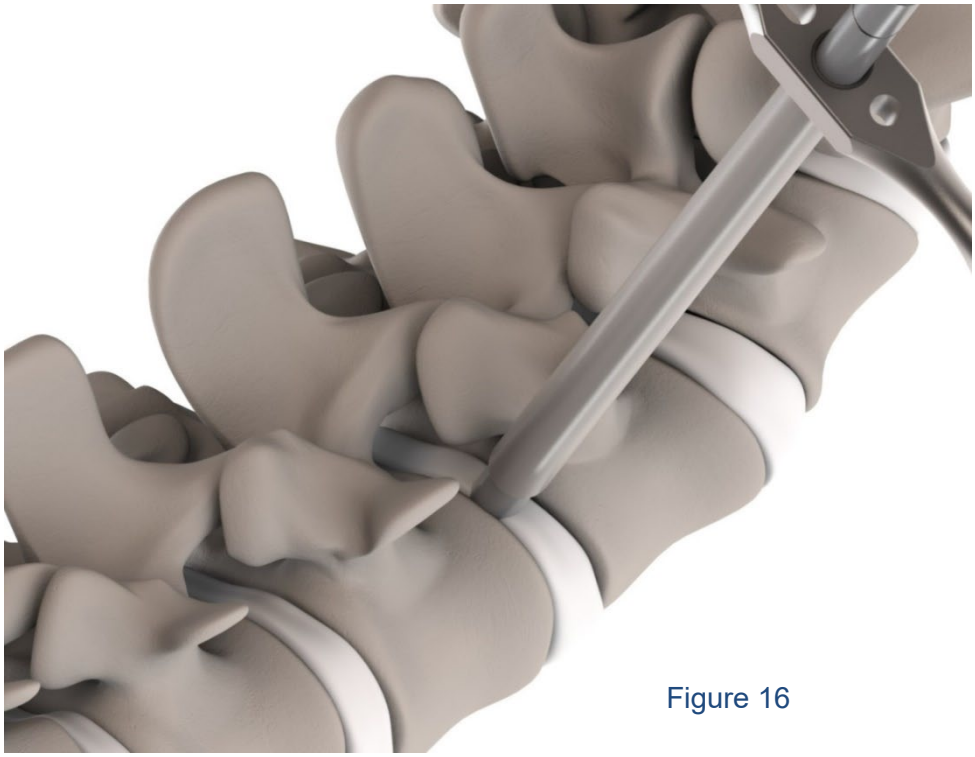


Figure 16

Place the **impactor** over the **dilator** and against the T of the **Access Portal** to provide an extension that can be used for the **mallet** to advance the **access portal**. **(Figure 17)**

Figure 17

Targeting

Remove the **impactor** and **dilator**, while controlling the **access portal**. It is critical to maintain control of the **access portal** at all times. One hand should always be holding the **access portal** and in contact with the skin to prevent a change in the depth of the **access portal** inside the disc space.

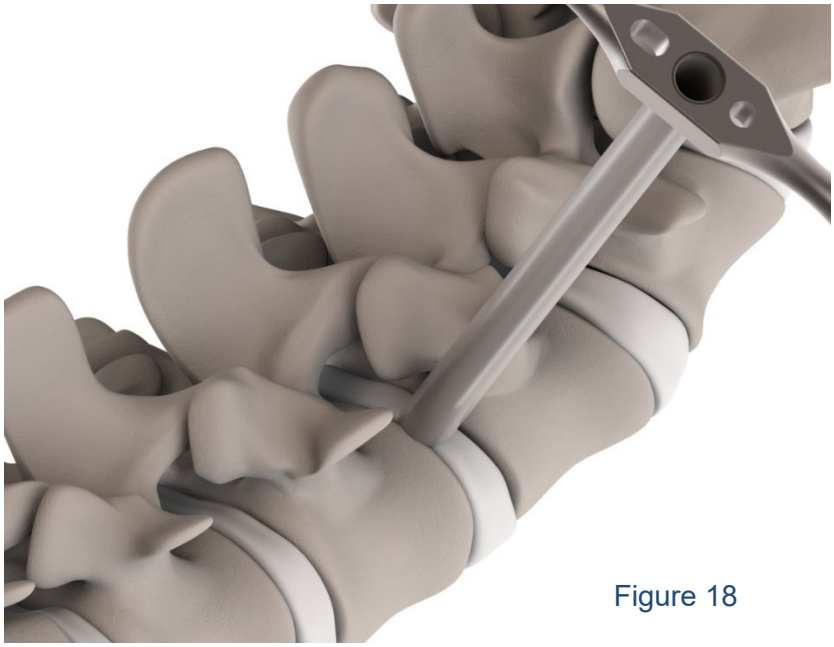


Figure 18

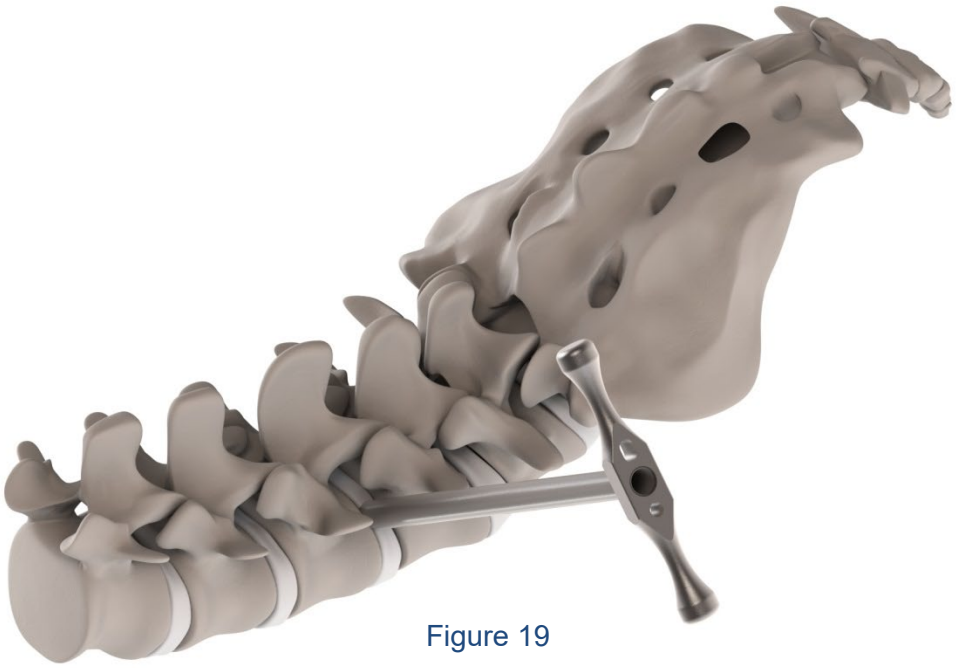


Figure 19

Discectomy

Under biplanar imaging, introduce the **8mm drill** through the **access portal** to remove a segment of the nucleus pulposus and prepare a space for further discectomy instruments.

Figure 20

The distance which the **8mm drill** protrudes past the cannula may be used as a method to determine the appropriate length of implant to be placed. Markings on the proximal shaft of the **drill** can be used to make this measurement.

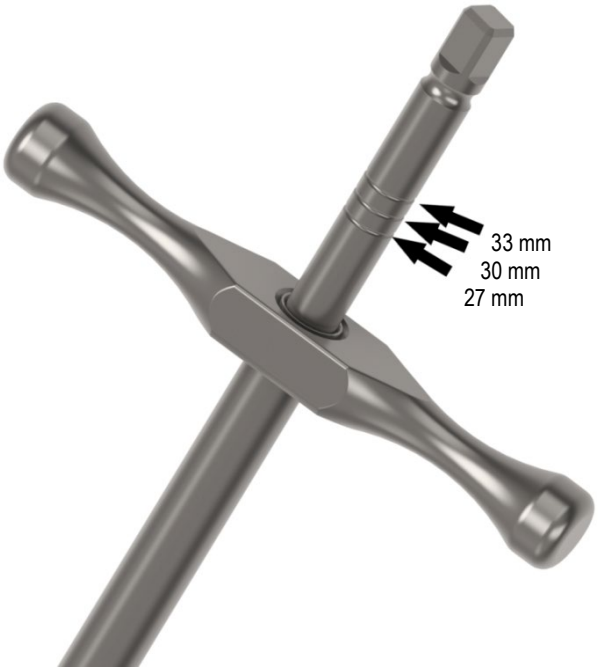


Figure 21

Discectomy

To prepare the endplate and debulk the nucleus of the disc, several instruments are included in the OLLIF instrument set.

<u>Shaper-</u> Rotate this instrument ONLY clockwise and expand in 1 mm increments to prepare the endplate. This instrument may be used to determine the height of the implant to be used. The window of the sliding sleeve of the body indicates the approximate height to which the **shaper blades** are expanded. This may be used as a guide for the height of the implant to best fit the disc space.

Figure 22

Discectomy

<u>Flexible Curette</u> – The nitinol blade of the **Flexible Curette** elastically regains it shape when introduced in the disc space and can be used in a fashion similar to a ring curette to scrape the endplate in a reciprocal (push/pull) motion.

A new blade should be used for each disc level prepared.

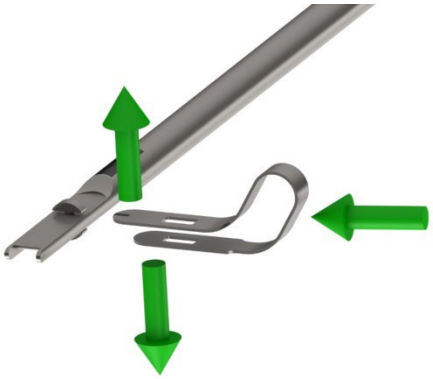
Figure 23

Discectomy

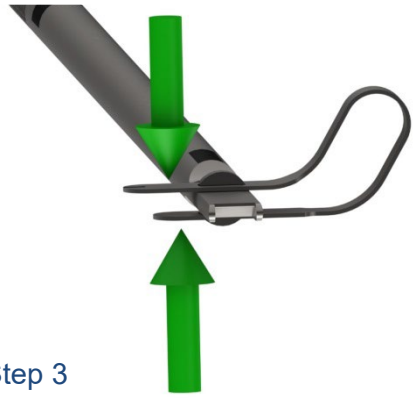
Preparing The Flexible Curette

Mount the single use **blade** on the **inner shaft** using the following method. One side of the **blade** is longer that the other. Align the longer side, which has a laser mark, so that it is on the same side of the shaft that also has a laser mark on it.

Hold the **blade** perpendicular to the axis of the inner shaft.

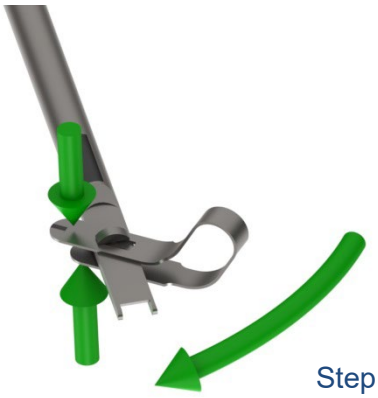


Spread the two arms of the blade



Step 3

Place the slots over the retainer posts on the top and bottom of the **inner shaft.**



Step 4

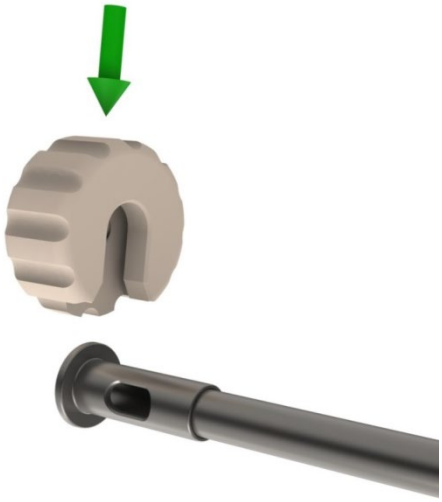
Press the blade arms into the **center shaft** and rotate the **blade** to align with the axis of the **inner shaft**. [Step 4 and Step 5]

Step 5

Step 6

Confirm that the **blade** is located between the side rails of the **inner shaft**.

Discectomy



Assemble the **Flexible Curette Knob** onto the **Flexible Curette Outer Sleeve** as shown.

Step 8

Place the quick connect end of the **Flexible Curette inner shaft** through the opening away from the flange of the **Flexible Curette outer sleeve**. Align the anti-rotation feature of the **inner shaft** with the window through the flange of the **Outer**.

Step 9

Guide the quick connect end of the **Flexible Curette Inner** through the **Sleeve** and out through the flange with the flat face of the quick connect being visible in the window.

Step 10

Continue to push the **shaft** into the **sleeve** while threading the **Knob** to connect with the threads on the **shaft.**

Discectomy

Connect the **Mini AO Quick Connect handle** when enough of the **Inner shaft** is exposed.

Continue to twist the **knob** in a clockwise direction to pull the **shaft** and **blade** into the **outer** until the laser etched alignment mark is visible inside the window of the flange.

This indicates that the **blade** is fully enclosed in the sleeve. Confirm that the blade is fully concealed at the opposite end.

To expose the **blade**, hold the **mini AO palm handle** and twist the **knob** in a counterclockwise direction to pull back the **sleeve** and expose the **blade**. When the second alignment mark is visible in the flange window, the **blade** is fully exposed. Confirm that the **blade** extends past the distal end of the instrument on the same side of the instrument as the window.

Discectomy

Surgical use

Before introducing the **Flexible Curette** to the **Access Portal**, twist the **knob** in a clockwise direction while holding the **mini AO palm handle** and retract the **shaft** into the **sleeve**, which will deform and conceal the superelastic nitinol **blade**.

Place the **Flexible Curette** assembly through the **Access Portal** and use the window in the flange to align the loop of the **blade** so that it will deploy in the disc space, exposing the edge of the loop perpendicular to the endplate. Twist the handle a few degrees toward one endplate until contact is felt with the bony surface of the endplate. Push and pull the **handle** to scrape the **blade** along the endplate of the vertebral body, scraping the cartilagenous surface of the endplate and exposing the cortical surface. Twist the **handle** in the opposite direction until is makes contact with the opposite endplate and repeat.
To addess the anterior section of the endplates, conceal the **blade** inside the sleeve and rotate the inner shaft to address the opposite direction.

To conceal the **blade**, hold the **handle** steady and twist the **knob** of the **Flexible Curette** in a clockwise direction to push the sleeve down and conceal the **blade.** Confirm that the alignment mark is visible in the flange window. Rotate the **Mini AO Quick Connect handle** 180 degrees and twist the **knob** in a counterclockwise direction to expose the **blade.**
Repeat the process to prepare the endplates on the anterior side of the disc space.

After completing the disc preparation using the **Flexible Curette**, hold the **Mini AO Quick Connect Handle** and twist the **knob** in a clockwise direction until the alignment mark is visible in the flange window. Withdraw the **Flexible Curette** from the **Access Portal**. Always maintain control of the **access portal** with one hand resting against the skin.
Use the **Pituitary Rongeur** to collect and remove fragments of disc that were scraped off of the endplate, exposing and preparing the cortical surface.

The extension from the axis of the flexible curette blade allows a wide pathway to be prepared. Because of this long reach, ARM does not recommend rotating the Flexible Curette inside the disc space. The distance from the axis of the instrument to the tip of the blade may be larger than the space between the endplates. Rotating the instrument while the blade is exposed places extreme stress on the instrument and may lead to a fracture of the attached blade.

Discectomy

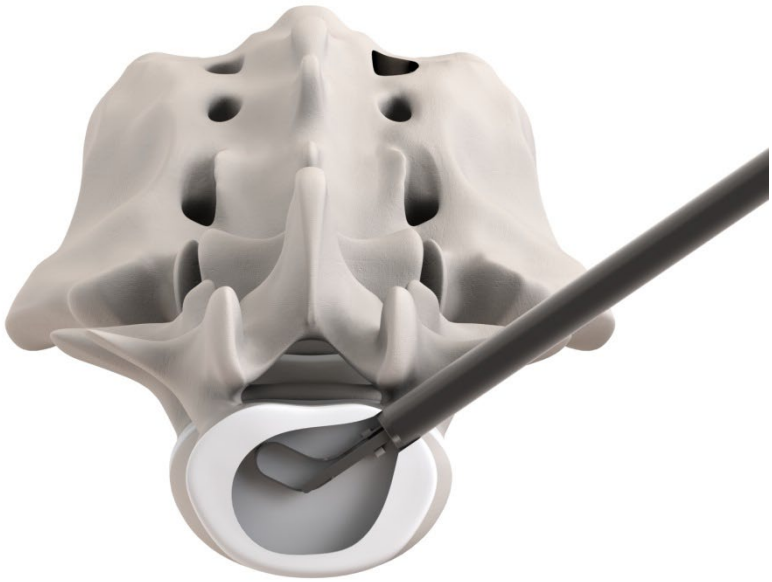


Figure 24

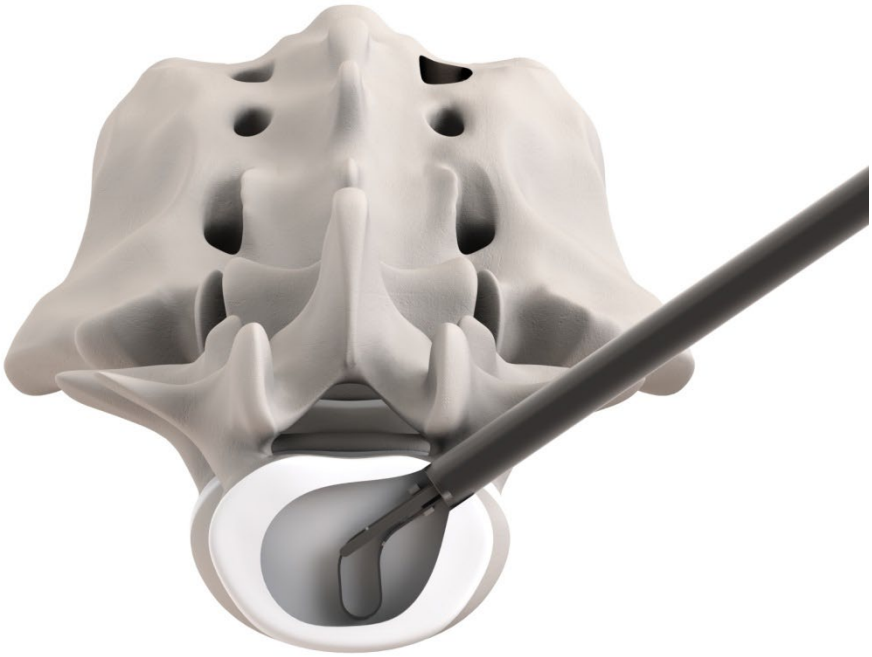


Figure 25

Discectomy

<u>Articulating Rake Curette</u> – this **curette** functions as an angled pull cup curette with the ability to adjust in height. Orient the flat surface to the endplate and place it distally in the disc and articulate the handles to push the flat rake tip against the endplate while pulling back toward the cannula.

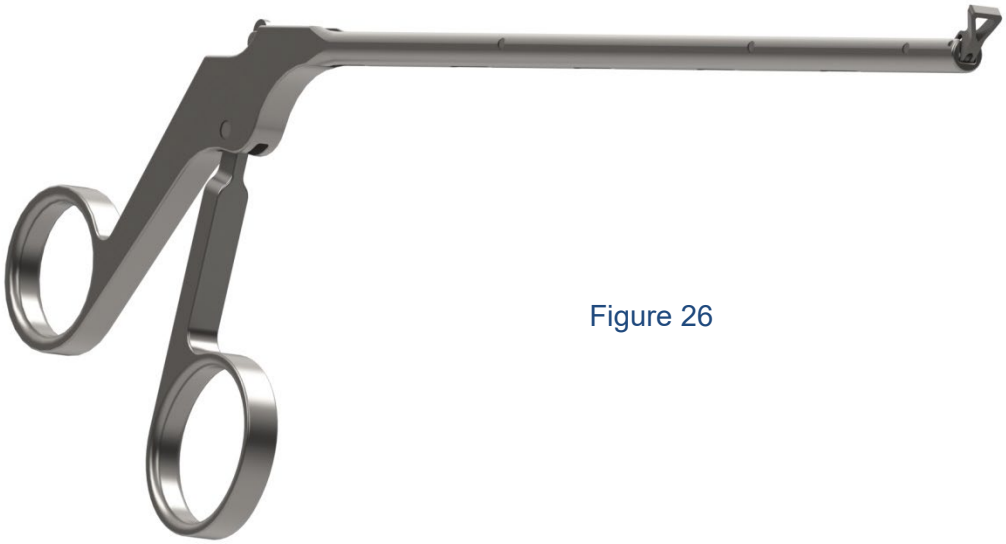
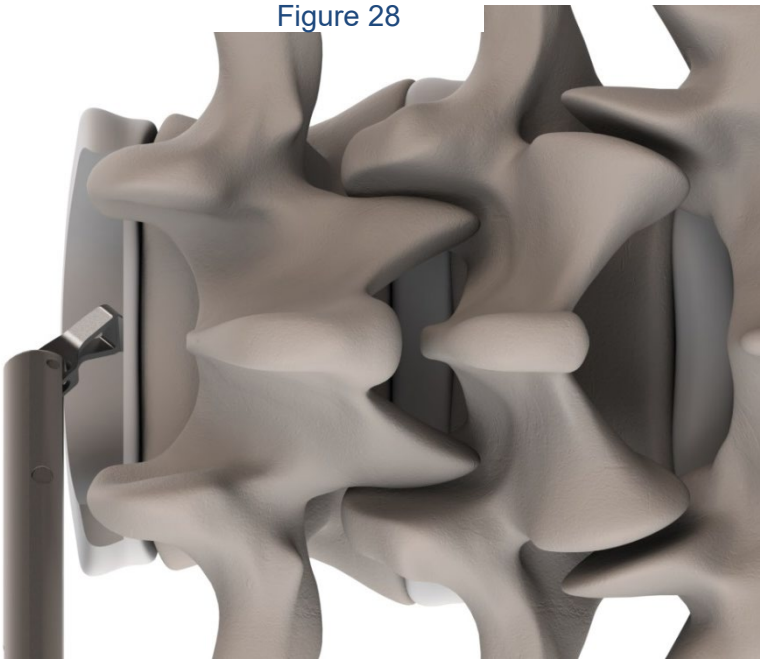


Figure 26

Figure 27

Figure 28



By using this combination of instruments and repeating the steps as necessary, a thorough discectomy may be achieved through the safe **Access Portal** which maintains the shielded pathway to the disc space.

Graft Delivery

A graft delivery system is included in the OLLIF instrument set to allow for delivery of graft material before the implant is introduced. The length of the **graft delivery sleeve** is matched to the length of the **access portal** and the **graft plunger** is matched to the sleeve.

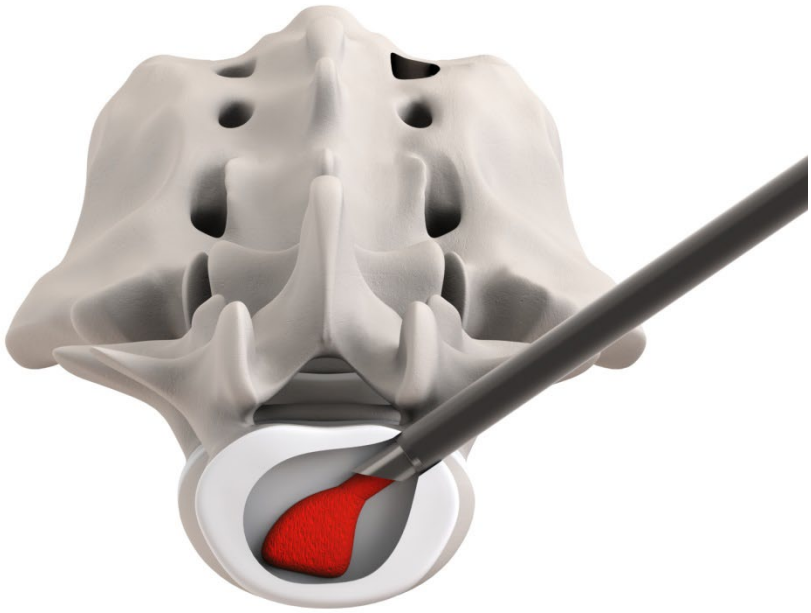


Figure 29

The beveled tip of the **sleeve** allows for delivery of the graft material in a direction chosen by the surgeon. An indicator on the flange indicates the orientation of the bevel tip of the **sleeve**.

Interbody Delivery

To complete the interbody delivery, place a **guide wire** through the **access portal** and into the graft mass inside the disc.

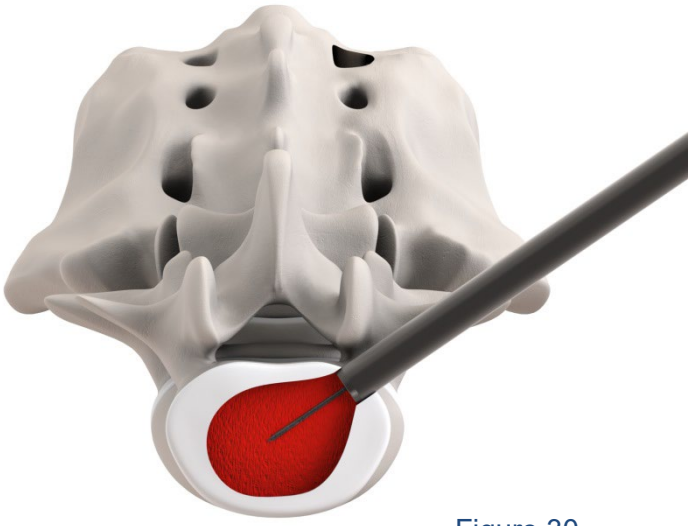


Figure 30

Carefully maintain the position of the **guide wire** and remove the **access portal**.

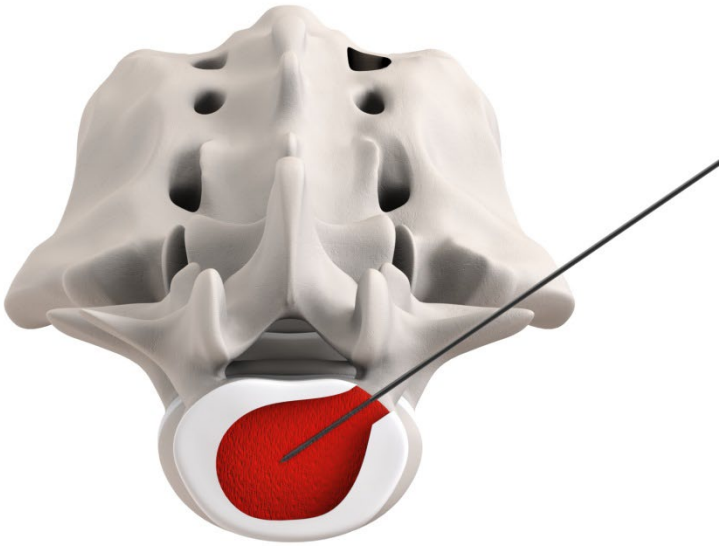


Figure 31

Interbody Delivery

The cannulated implant and **inserter** are placed over the **guide wire** to follow the safe path created into the center of the disc.

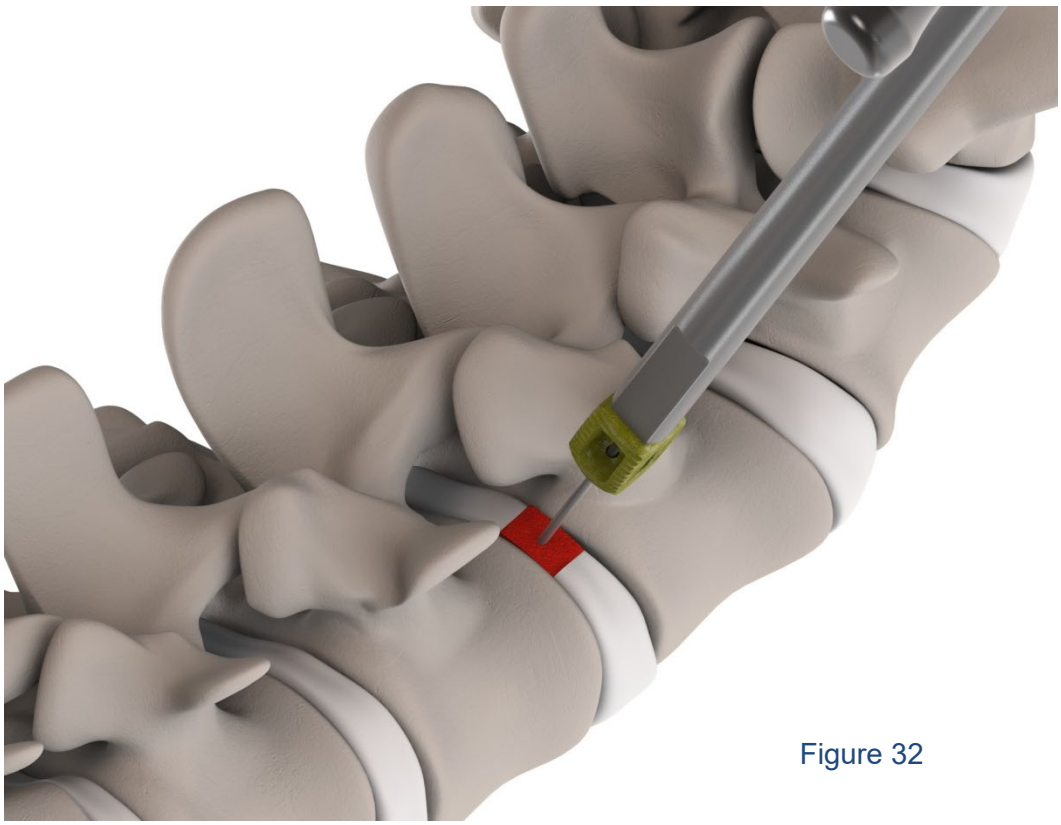


Figure 32

Use the **mallet** to impact the **inserter** and place the graft securely in the center of the disc. The location of the titanium implant can be easily determined through fluoroscopy. The orientation of the interbody's endplate surface is perpendicular to T-handle of the **inserter**.

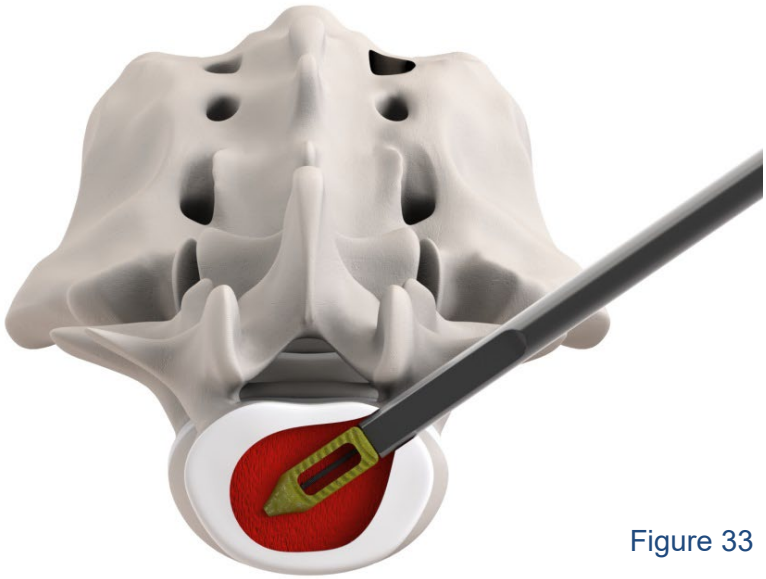


Figure 33

Interbody Delivery

To release the implant from the **inserter**, turn the retainer **knob** in a counter clock-wise fashion until the implant is not connected to the **inserter** and withdraw the **inserter.** A **wrench** is included to add leverage and grip at this point.

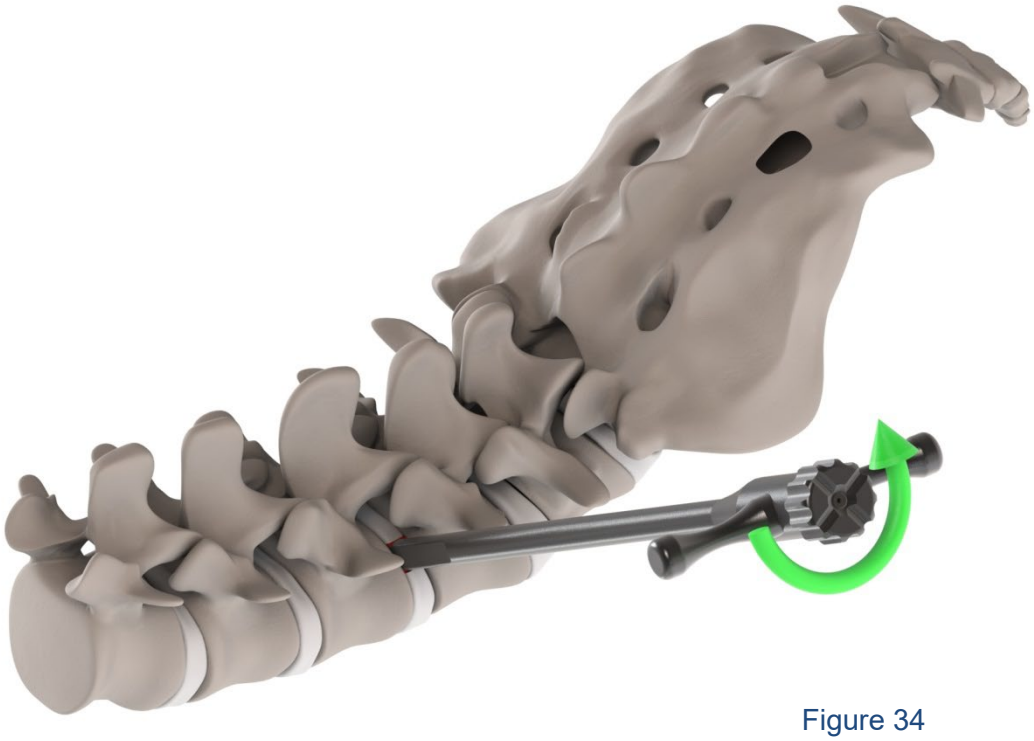


Figure 34

Removal

In the event that an implant needs to be removed after placement, follow the same surgical approach with a **neuromonitoring probe** and **guide wire** to place the **dilator** through Kambin's Triangle. Use the **dilator** to locate the pocket on the back of the interbody device and pass the **guide wire** through the **dilator** and into the threaded aperture on the interbody device.

Remove the **dilator** and place only the OLLIF **inserter inner shaft** over the **guide wire** to connect to the threads of the implant. Thread the **inner shaft** into the implant as far as possible.

Using a **mallet**, tap the **inner shaft** to withdraw the implant.

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